

2.1 Integration by Parts

General Idea:

We want to use the product rule to develop a useful formula for integration.

Recall that for differentiable functions f and g the product rule is:

$$\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + f'(x)g(x)$$

Now, using our knowledge of indefinite integration we obtain:

$$\int [f(x)g'(x) + f'(x)g(x)] dx = f(x)g(x) + C$$

$$\Leftrightarrow \int f(x)g'(x) dx + \int g(x)f'(x) dx = f(x)g(x) + C$$

$$\Leftrightarrow \int f(x)g'(x) dx = f(x)g(x) - \int g(x)f'(x) dx$$

Note: In the last line the constant C is no longer written since the indefinite integral on the right will be a function of x plus a constant.

Now, if we let $u = f(x)$ and $v = g(x)$, then $du = f'(x)dx$ and $dv = g'(x)dx$.

Using this notation the above equation becomes:

$$\int u dv = uv - \int v du$$

As it turns out this formula can be quite useful for integrating certain functions!

Formula (Integration by parts):

$$\int u dv = uv - \int v du$$

Example 1:

(a) Find $\int x \cos(x) dx$.

One of the most challenging aspects of using integration by parts is deciding which part of the integral to let u equal and which part of the integral to let dv equal. Roughly speaking it is a good idea to let u equal something that becomes “nicer” when differentiated, and let dv equal something that can be easily integrated.

In this case let:

$$u = x \quad \& \quad dv = \cos(x) dx \quad \text{then,} \quad du = dx \quad \& \quad v = \sin(x)$$

So using the integration by parts formula we obtain:

$$\int x \cos(x) dx = x \sin(x) - \int \sin(x) dx = x \sin(x) - [-\cos(x)] + C = x \sin(x) + \cos(x) + C$$

(b) If you let $u = \cos(x)$ & $dv = x dx$ what does the integration by parts formula yield?

We let: $u = \cos(x)$ & $dv = x dx$

These two equations imply: $du = -\sin(x) dx$ & $v = \frac{1}{2}x^2$.

So using the integration by parts formula we obtain:

$$\int x \cos(x) dx = \frac{1}{2}x^2 \cos(x) - \int -\frac{1}{2}x^2 \sin(x) dx$$

*Note: that the integral on the right hand side of the equality is more complicated than the original integral!
This illustrates the importance of picking u and dv .*

Example 2:

Find $\int \ln(x) dx$

We use integration by parts and we let: $u = \ln(x)$ & $dv = dx$.

These two equations imply: $du = \frac{1}{x} dx$ & $v = x$.

So using the integration by parts formula we obtain:

$$\int \ln(x) dx = x \ln(x) - \int x \left(\frac{1}{x} \right) dx = x \ln(x) - \int 1 dx = x \ln(x) - x + C$$

Question: How could we check our answer here to make sure it is correct?

Note: Sometimes we need to apply the integration by parts formula more than one time.

Example 3:

Find $\int x^2 \sin(x) dx$

For this problem we let: $u = x^2$ & $dv = \sin(x)$.

These two equations imply: $du = 2x dx$ & $v = -\cos(x)$.

So using the integration by parts formula we obtain:

$$\int x^2 \sin(x) dx = -x^2 \cos(x) - \int -2x \cos(x) dx = -x^2 \cos(x) + 2 \int x \cos(x) dx$$

We use integration by parts again.

Let $u = x$ & $dv = \cos(x) dx$.

Then: $du = dx$ & $v = \sin(x)$.

Now we have:

$$\begin{aligned} \int x^2 \sin(x) dx &= -x^2 \cos(x) + 2 \int x \cos(x) dx = -x^2 \cos(x) + 2 [x \sin(x) + \cos(x) + C] \\ &= -x^2 \cos(x) + 2x \sin(x) + 2 \cos(x) + C' \end{aligned}$$

Note: We use a new constant C' which is the same thing as two times C .

Theorem (Tabular Integration by Parts): Suppose $p(x)$ is a polynomial of degree n where n is a natural number, and suppose that $F_0(x), F_1(x), \dots, F_{n+1}(x)$ are functions such that $F_{i+1}(x)$ is an antiderivative of $F_i(x)$ for each $0 \leq i \leq n$. We have that:

$$\int p(x) F_0(x) dx = \sum_{i=0}^n (-1)^i p^{(i)}(x) F_{i+1}(x) + C$$

Where $p^{(i)}(x)$ is the i^{th} derivative of $p(x)$.

Note: This theorem is not emphasized in the book, but it is quite useful.

Proof (not done in class):

We will show that the first derivative of $\sum_{i=0}^n (-1)^i p^{(i)}(x) F_{i+1}(x) + C$ is $p(x) F_0(x)$ which will complete the proof.

Now, we use the product rule and simplify to obtain:

$$\begin{aligned} \frac{d}{dx} \left(\sum_{i=0}^n (-1)^i p^{(i)}(x) F_{i+1}(x) + C \right) &= \sum_{i=0}^n \left[\frac{d}{dx} \left((-1)^i p^{(i)}(x) F_{i+1}(x) \right) \right] \\ &= \sum_{i=0}^n \left((-1)^i p^{(i+1)}(x) F_{i+1}(x) + (-1)^i p^{(i)}(x) F_i(x) \right) \\ &= \sum_{i=0}^n (-1)^i p^{(i+1)}(x) F_{i+1}(x) + \sum_{i=0}^n (-1)^i p^{(i)}(x) F_i(x) \\ &= \sum_{i=1}^{n+1} (-1)^{i-1} p^{(i)}(x) F_i(x) - \sum_{i=0}^n (-1)^{i-1} p^{(i)}(x) F_i(x) \\ &= (-1)^n p^{(n+1)}(x) F_{n+1}(x) - \left((-1)^{-1} p(x) F_0(x) \right) \\ &= (-1)^n p^{(n+1)}(x) F_{n+1}(x) + p(x) F_0(x) \end{aligned}$$

Now, since $p(x)$ is a polynomial of degree n we know $p^{(n+1)}(x) = 0$. Thus, we have:

$(-1)^n p^{(n+1)}(x) F_{n+1}(x) + p(x) F_0(x) = p(x) F_0(x)$, and we are finished. ■

Note: This proof can also be done by repeatedly applying the integration by parts formula.

Visualizing Tabular Integration by Parts:

Consider the integral: $\int x^2 \sin(x) dx$. One way to apply the above theorem is to use a table:

x^2	$\sin(x)$	
x^2	$\sin(x)$	
$2x$	$-\cos(x)$	+
2	$-\sin(x)$	-
0	$\cos(x)$	+

Following the arrows of the table we obtain a solution of: $-x^2 \cos(x) + 2x \sin(x) + 2 \cos(x) + C$.

Note: This solution agrees with what we found in the most recent example! Also, one can view the third column as alternating between positive and negative one. So, to get the first term we did x^2 times $-\cos(x)$ times 1. To get the second term we did $2x$ times $-\sin(x)$ times -1. Finally, to get the third term we did 2 times $\cos(x)$ times 1.

Example 4 (you try):

Find $\int (x^2 + 1)e^{-x} dx$.

Using tabular integration, we obtain:

x^2+1	e^{-x}	
x^2+1	e^{-x}	
$2x$	$-e^{-x}$	+
2	e^{-x}	-
0	$-e^{-x}$	+

So, the solution is: $\int (x^2 + 1)e^{-x} dx = -(x^2 + 1)e^{-x} - 2xe^{-x} - 2e^{-x} + C = -x^2e^{-x} - 2xe^{-x} - 3e^{-x} + C$

Formula (Integration by Parts for Definite Integrals): If f' and g' are continuous on $[a, b]$, then

$$\int_a^b f(x) g'(x) dx = f(x) g(x) \Big|_a^b - \int_a^b g(x) f'(x) dx$$

Example 5:

Calculate the exact value of: $\int_1^2 [\ln(x)]^2 dx$

Let $u = [\ln(x)]^2$ & $dv = dx$. These two equations imply $du = \frac{2 \ln(x)}{x} dx$ & $v = x$.

Now, if we apply the integration by parts formula we obtain:

$$\begin{aligned} \int_1^2 [\ln(x)]^2 dx &= x [\ln(x)]^2 \Big|_1^2 - \int_1^2 x \left(\frac{2 \ln(x)}{x} \right) dx \\ &= 2 [\ln(2)]^2 - [\ln(1)]^2 - 2 \int_1^2 \ln(x) dx = 2 [\ln(2)]^2 - 2 \int_1^2 \ln(x) dx \end{aligned}$$

We again use integration by parts:

Let $u = \ln(x)$ & $dv = dx$.

Then $du = \frac{1}{x} dx$ & $v = x$.

Therefore we have:

$$\begin{aligned} \int_1^2 [\ln(x)]^2 dx &= 2 [\ln(2)]^2 - 2 \int_1^2 \ln(x) dx \\ &= 2 [\ln(2)]^2 - 2 \left[x \ln(x) \Big|_1^2 - \int_1^2 x \left(\frac{1}{x} \right) dx \right] \\ &= 2 [\ln(2)]^2 - 2 (2 \ln(2) - 1) = 2 [\ln(2)]^2 - 4 \ln(2) + 2 \end{aligned}$$

Case of the Unknown Integral

Example 6:

Find $\int e^x \cos(x) dx$

We let $u = \cos(x)$ and $dv = e^x dx$.

These two equations imply: $du = -\sin(x)dx$ and $v = e^x$.

Now, we apply the integration by parts formula and we obtain:

$$\int e^x \cos(x) dx = \cos(x)e^x - \int e^x (-\sin(x)) dx = \cos(x)e^x + \int e^x \sin(x) dx.$$

We again use integration by parts:

Let $u = \sin(x)$ and $dv = e^x dx$.

Then $du = \cos(x)dx$ and $v = e^x$.

Therefore we have:

$$\begin{aligned} \int e^x \cos(x) dx &= \cos(x)e^x + \int e^x \sin(x) dx \\ &= \cos(x)e^x + \left[\sin(x)e^x - \int e^x \cos(x) dx \right] \end{aligned}$$

We notice that the original starting integral has reappeared on the RHS and so we solve for it.

$$\begin{aligned} \int e^x \cos(x) dx &= \cos(x)e^x + \sin(x)e^x - \int e^x \cos(x) dx \\ \Leftrightarrow 2 \int e^x \cos(x) dx &= \cos(x)e^x + \sin(x)e^x \\ \Leftrightarrow \int e^x \cos(x) dx &= \frac{1}{2} [\cos(x)e^x + \sin(x)e^x] + C \end{aligned}$$